

LD-Aware Attention for Genomic Analysis

Fill in missing genotypes without building an r^2 table

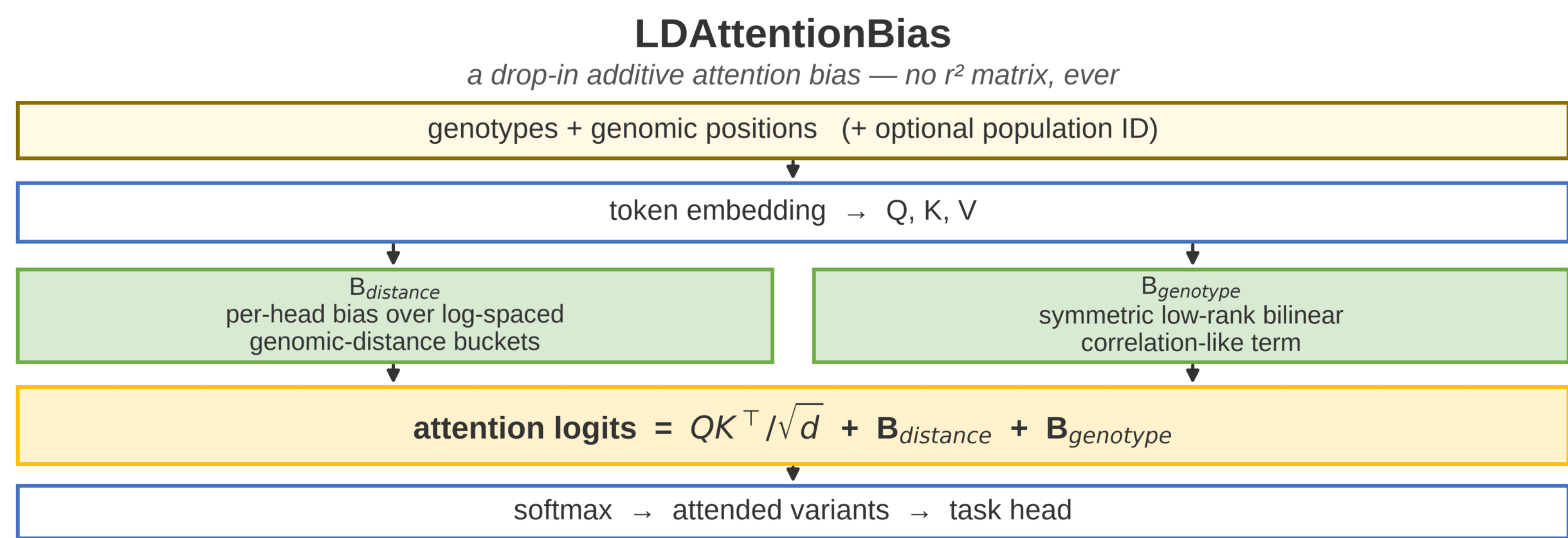
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1 Background

Nearby DNA variants are often inherited together. That non-random pairing is called linkage disequilibrium (LD). Geneticists measure it with r^2 : a score from 0 (independent) to 1 (always travel together). Standard pipelines precompute every pairwise r^2 and store an $L \times L$ table, then rebuild that table for every new cohort, window and frequency cutoff.

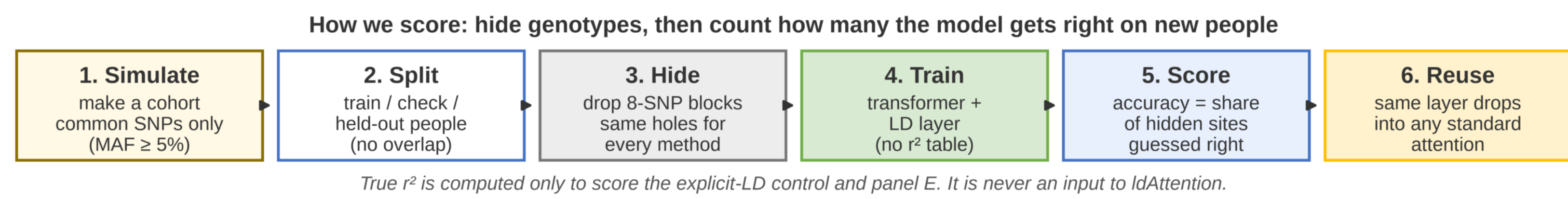
Imputation is guessing a genotype the lab did not observe. When neighbouring SNPs drop out together — as they do in genotyping-by-sequencing — a stored list of LD partners vanishes with the site it was meant to rescue. IdAttention is a small add-on that teaches attention that LD structure, so it can still look across the rest of the window, without ever building the r^2 table.

2 Approach



- **What it does:** Before attention picks which variants to look at, we add two learned numbers — an LD bias — to those scores.
- **Distance term:** Nearby sites get a boost. LD fades with genomic distance; this term learns that fade.
- **Genotype term:** Sites whose alleles move together get a boost — a correlation-like pairwise score, learned from the data.
- **Where it sits:** A drop-in for any standard attention layer (PyTorch SDPA or MultiheadAttention). No extra preprocessing stage.
- **What it removes:** The stored $L \times L$ r^2 table. Nothing to build, store, or rebuild when the cohort or the window changes.

3 Methods



128 SNPs $\times$ 1000 people (msprime, MAF $\geq 5\%$). Hide 30% of genotypes in 8-SNP blocks (as in GBS). Accuracy = % guessed right on held-out people. A point = 1 pp of that. Same holes, 6 seeds.

4 Results

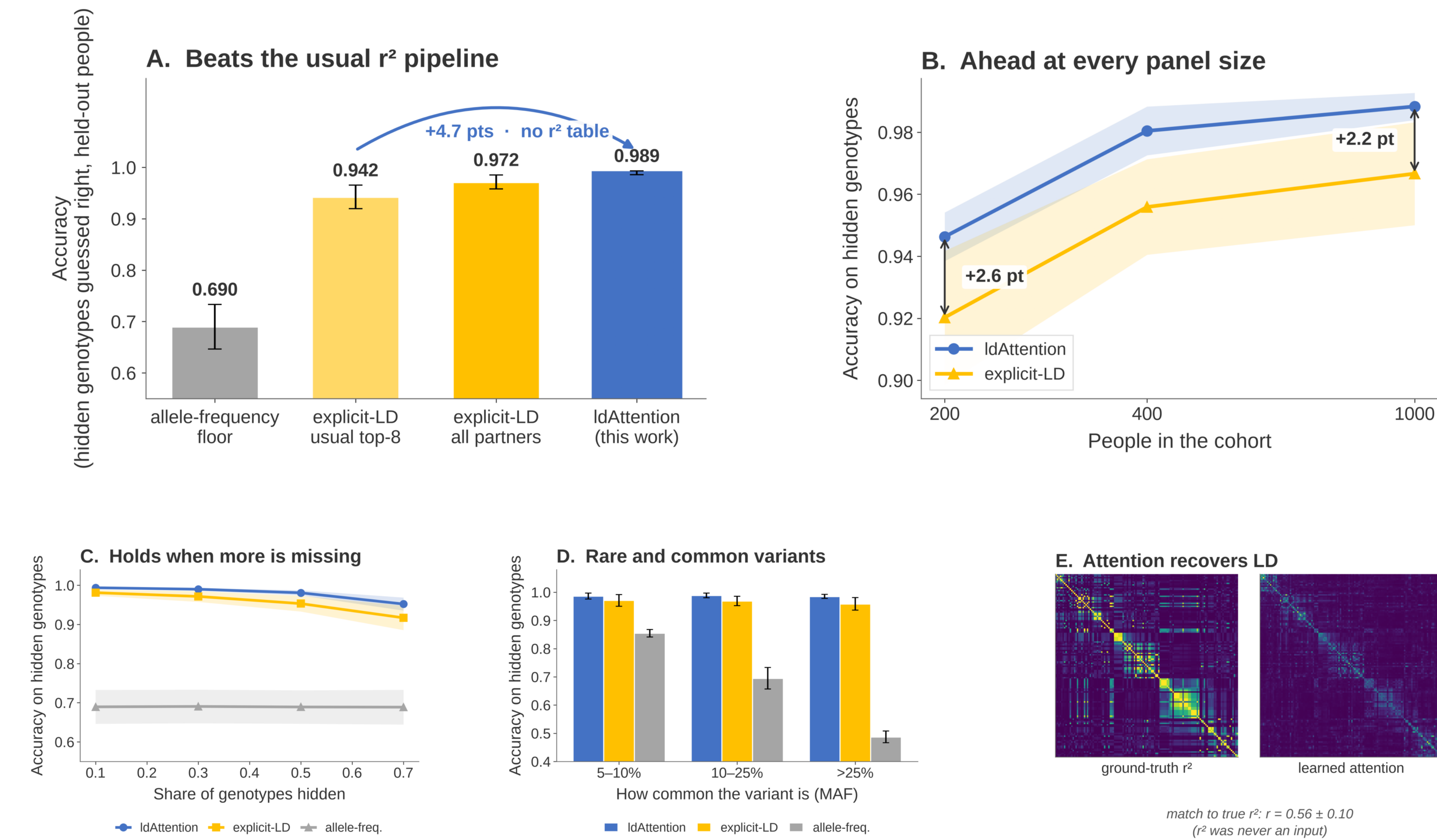


Figure 1. Accuracy = % hidden genotypes right on held-out people; a point = 1 pp. a) vs usual and fully-tuned r^2 pipelines. b) ahead of explicit LD at every cohort size. c) More genotypes hidden. d) By how common the variant is (MAF). e) True r^2 vs attention. Mean $\pm$ s.d., 6 seeds (a, c, d); 3 (b).

5 Conclusion

- **Beats explicit LD:** 98.9% of hidden genotypes right, +4.7 pts vs the usual r^2 pipeline.
- **Recovers LD:** attention matches the true r^2 table ($r = 0.56$), never shown it.
- **Nothing to precompute:** no stored $L \times L$ table, no rebuild when the cohort or window changes.

Accuracy = % of hidden genotypes right on new people; a point = 1 pp of that. The layer's edge: beat the r^2 pipeline without building that table (a) and reconstruct the table in attention (e). Fully-tuned all-partner control: 97.2% (+1.8 pts).



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Code & pre-print
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